PWCS ACT CONFERENCE · HIGH SCHOOL SCIENCE

From Learning Target to iPad

Creating custom CER readings with AI

Biology · Chemistry · AP · Earth Science

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October 12, 2026 · Hylton High School
Slides: scienceonepagers.org/cer-slides

Before 0:00 AS YOU SIT DOWN

Log in, pick a target, pair up

1

Log in to ChatGPT

Your PWCS account. Start a new chat and choose GPT-5.6 Sol.

2

Pick a target you teach this fall

Biology: the list on your table. Chemistry, AP, earth science: your own.

3

Sit with a partner

You trade drafts at the end. Any course works.

Wi-Fi: DIIT, automatic · PWCS Reg. 295-5: approved tools, human judgment, protect privacy, be transparent, verify the output

0:00 · 15 min BE THE STUDENT FIRST

Read, talk, write a finished CER

Read · 7 Pick one reading. Mark it as you go.

Talk · 4 With a partner: what is the claim, and what would you cite?

Write · 4 One claim and two pieces of evidence.

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Chesapeake · Bear · Heat and enzymes · Chronic wasting disease · Lake Anna · Elephants

0:15 · 5 min

THE PROTOCOL YOU JUST RAN

Read · Talk · Write

Read

Alone, pencil moving. Mark claims to TRACE and words to FLAG.

Talk

With a partner, out loud: "I think… because the reading says…"

Write

The CER, alone, after the talk. Talk is the rehearsal.

Talking first is what makes the writing possible.

0:20 WHY WE DO THIS

Two articles to read while you build

Read, Talk, Write

The protocol, the research, the rubric

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CER readings with ChatGPT

Every prompt and the four checks

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A reading tied to what we are teaching keeps students engaged, helps them hold on to it, and builds critical thinking. It is not an SOL score booster.

0:20 FROM TARGET TO STUDENT

One target in, one reading out

THE LEARNING TARGET

LT 2.5 Cell Transport

Students can explain how materials move across the cell membrane and predict which way they will move.

→

Road Salt, Roadside Plants, and the Movement of Water Across Cell Membranes

Each winter, trucks spread salt on roads across Northern Virginia, including roads in Prince William County. When snow and ice melt, some of the salty water flows into the soil beside the road…

Routine: Read · Talk · Write | Delivery: a Word file on the student iPad

0:20 WHO SETS THE RIGOR

The teacher sets the rigor, not the AI

A bare prompt

"Write a reading about cells."

ChatGPT chooses the course, the level, the vocabulary, the place and the standard. You get a list of facts to retrieve.

A scaffolded CER

Claim · why the roadside plants wilt

Evidence · what the reading shows about salt in the soil

Reasoning · osmosis, the rule that connects them

Goal: bridge the gap between running a procedural test and understanding why the mechanism works.

0:22 · 3 min TAKE IT APART

What that page is made of

A real phenomenon

One place, person or measured value, close to home.

The reading

Only words your students have already met.

The evidence

Real quantities with units, from sources students can check.

The CER task

Each question tied to its SOL or AP objective.

0:22 WHAT MAKES IT WORK

A reading worth talking about

One page. About 600 to 750 words, long enough to hold evidence.

A real question. Something students could disagree about.

Numbers to argue over. Evidence a student can point at.

A place they know. A stake close to home.

TWO RULES GOVERN EVERY LINE

1 **No term students have not met.**

2 **Every claim traces to a source.**

In an AP course the vocabulary list is longer. The rule is the same.

0:25 · 5 min WHAT A STRONG CER LOOKS LIKE

The reading has to make a 3 possible

	3 · Meets	2 · Developing	1 · Beginning
Claim	Accurate, complete, in science terms	Partly accurate or vague	Inaccurate or missing
Evidence	Enough relevant data from the reading	Some, but thin	Little or none
Reasoning	Explains why the evidence supports the claim	Connects, but the science is weak	No link, or a misconception

No numbers in the reading, no 3 in Evidence. That is why the scaffold asks for three.

0:28 SHORT ON CLASS TIME

Split the protocol across the week

AT HOME

Read

One page on the iPad the night before. A little reading at home is fine.

IN CLASS

Talk

Five to eight minutes with a partner, while you listen in.

IN FLEX

Write

The CER in the Word file, finished and turned in.

0:30 PWCS BIOLOGY · YEAR AT A GLANCE

Units 3–9 have no reading yet

1 · Biochemistry

1.1 **1.2** **1.3**

2 · Cells

2.1 **2.2** **2.3** 2.4 2.5

3 · Cell Energetics

3.1 3.2 3.3 3.4

4 · Nucleic Acids

4.1 4.2 4.3 4.4

5 · Growth & Division

5.1 5.2 5.3 5.4

6 · Genetics

6.1 6.2

7 · Evolution

7.1 7.2 7.3

8 · Classification

8.1 8.2 8.3

9 · Ecology

9.1 9.2 9.3 9.4

5 of 32 targets have a published reading. **27 are open.**

Filled = published CER reading on scienceonepagers.org · checked October 11, 2026

0:30 · 25 min YOUR TURN

The room becomes a lab

1. Open ChatGPT on your PWCS account
2. Pick a target you teach in October
3. Run the prompt from your handout
4. Check it: TRACE, FLAG, COUNT, MATCH
5. Paste it into Word for the iPad

STUCK? BORROW A MODEL

About fifty finished one-pagers

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Work in pairs, side by side | GPT-5.6 Sol, new chat

0:30 · 5 min CREATE · PASS 1 OF 3

Choose the phenomenon by hand

Name one specific place, person or measured value your students could picture.

WHY BY HAND

Left alone, ChatGPT opened on "Each winter, trucks spread salt." That is a generalization, not a hook.

0:30 · 5 min CREATE · PASS 1 OF 3

Close to home, in every course

Biology · LT 2.5 Cell Transport

Road salt on Northern Virginia roads

Chemistry · Gas laws

The tire-pressure light on a cold October morning

AP Biology · Evolution

Who can still digest milk as an adult, and why

Earth Science · Water

The Occoquan Reservoir after a heavy rain

0:35 · 10 min CREATE · PASS 2 OF 3

Type this to get started

Help me, a high school [course] teacher in Prince William County, Virginia, write a one-page student reading with a claim-evidence-reasoning task. Learning target: [target]. Phenomenon: [a place, person or measured value near here]. Before you write, ask me what vocabulary my students have met and what they have not studied yet.

Full scaffold: your handout or scienceonepagers.org/cer · new chat, GPT-5.6 Sol

Used only this starter? Follow up: "Add three real numbers with units and a list of sources to verify."

0:35 · 10 min CREATE · PASS 2 OF 3

Inside the full scaffold

1 · ROLE AND RULES

Help a PWCS teacher. No term off the list. Nothing from a later unit. Every claim checkable.

2 · THE BLANKS YOU FILL

Target and SOL or AP code · phenomenon · vocabulary met · not yet studied · the trap

3 · THE SHAPE

Opening names a place or number · three sections with word budgets · CER task · sources · uncertain

Method: one tested prompt on your PWCS account | Handout: the full prompt, ready to fill in

0:35 · 10 min WHAT WENT IN, WHAT CAME OUT

One prompt, one reading, one standard

WENT IN

TARGET: LT 2.5 Cell Transport

PHENOMENON: road salt, roadside plants, streams

RULE: no ATP; say energy

RULE: list SOURCES TO VERIFY

CAME OUT, MARKED

Scientists have measured higher amounts of salt in many streams **[TRACE]**

Water moves … to where there is less available water **[MATCH]**

[FLAG] none: the vocabulary fence held

THE ANCHOR

LT 2.5, Unit 2 Cells

Membrane, transport, homeostasis: BIO.3

Claims checked against sources: BIO.1

Method: a scaffolded prompt | Delivery: a Word file on the student iPad

0:45 · 10 min CREATE · PASS 3 OF 3

Check what came back

TRACE
Every claim to a source you could open.

FLAG
Every term your students have not met.

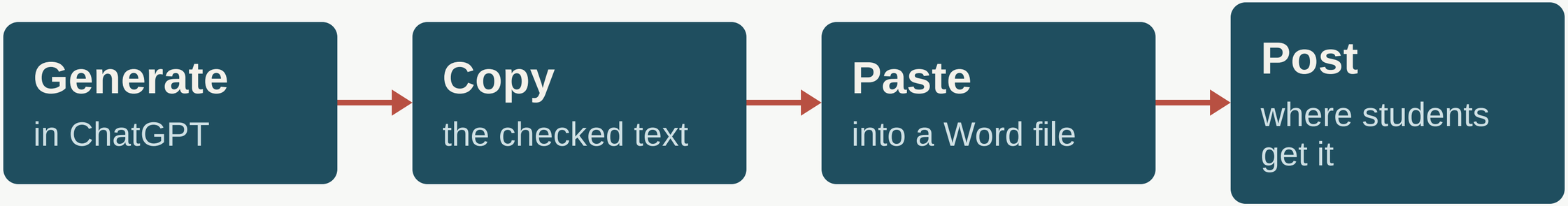
COUNT
Three real numbers, each with a unit.

MATCH
The key definition, every time it appears.

Students run the same four checks. That is the ownership.

0:55 · 5 min GET IT TO THE STUDENT

Four moves, all by hand



Division AI does not reach the browser or Canvas, so you can repeat every step Monday with nothing else installed.

0:55 · 5 min GET IT TO THE STUDENT

What works on the iPad

Works

- A Word file students type into; it saves as they go
- Several students in one shared file
- The finished file uploaded to Canvas

Quietly fails

- A PDF where students need to type
- Student Annotation: it seals each attempt
- Inserting a chart in Word on the iPad
- Google formats on the student device

1:00 · 7 min TRADE DRAFTS

Swap drafts and find one error

Run the same four checks on your partner's draft.

Mark one thing that is wrong, and show them where.

Partner teaches a different course? Good. You read it the way a student who has not met the vocabulary would.

A REAL DRAFT · BIOLOGY LT 2.5 · SEPTEMBER 13

The error MATCH caught

> Diffusion: "from an area of higher concentration to an area of lower concentration."
>
> Osmosis: water moves "from where there is more water to where there is less available water."

Concentration of what? A student tracking carefully gets two different answers, and the draft passed every other rule.

Accuracy report

BIOLOGY LT 2.5 · ROAD SALT · DIVISION ACCOUNT

What the four checks found

	Sept 13 · Instant · full prompt	Oct 11 · Sol · short starter	Oct 11 · Sol · full scaffold
TRACE	Real sources named, nothing invented	**No sources; made-up class data**	Seven claims, each with a named source
FLAG	Fence held; never said ATP	**"Cytoplasm" used, not on the list**	Fence held; "chloride" defined in plain words
COUNT	**No numbers at all**	**No numbers; High, Medium, Low**	Four quantities with units
MATCH	**Diffusion and osmosis disagree**	One definition of osmosis throughout	One definition; the trap named outright

Every run needed the checks. A listed source counts only after someone opens it.

1:07 · 3 min

SHARE WHAT YOU BUILT

Add yours to a shared CER library

1. Save it as a Word file
2. Put your initials and school first: **RS_BFHS**_your title
3. Drop it in **_Drop your CER here**

Using someone else's? Save a copy first, then adapt it.

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1:10 · 4 min YOUR TURN TO TEACH IT BACK

Where would RTW fit in your week?

Talk: tell your partner which class, which day, and which part happens at home, in class, or in FLEX.

Write: the same answer in the form. We read every one.

Feedback form, two minutes

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1:14 WHERE TO GET EVERYTHING

Two links to keep

The prompts

Every prompt and the four checks, ready to copy

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A shared library

Share your CER and use everyone else's

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These slides: **scienceonepagers.org/cer-slides** · Reading: **scienceonepagers.org/rtw** · Reid Schwebach · Katie Novinger